



Cambridge IGCSE™

COMBINED SCIENCE

Paper 6 Alternative to Practical

MARK SCHEME

Maximum Mark: 40

0653/61

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This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **12** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	correct awarding one mark from marking point or marking group 1 similar numbered ticks are used for marking point or marking groups 2, 3, 4 and 5
	information missing or insufficient for credit
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	response is too vague or there is insufficient detail in response
	incorrect point or mark not awarded

Annotation	Meaning
	point has been noted, but no credit has been given or blank page seen
	key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	blank page
	unclear response
	two statements are linked
	follow through
	irrelevant material that does not answer the question

Question	Answer	Marks	Guidance
1(a)(i)	21.5 ; 24.0 ;	2	
1(a)(ii)	0.0 and 3.5 ;	1	Allow 0 ECF from 1(a)(i)
1(a)(iii)	vertical axis labelled <i>change in temperature</i> and °C AND horizontal axis labelled <i>percentage concentration of hydrogen peroxide</i> ; suitable linear scales so that points occupy more than half the grid ; plots correct $\pm$ half a small square ;	3	Allow DT / °C Allow concentration of hydrogen peroxide / % Ignore percentage concentration / concentration of hydrogen peroxide alone Do not allow C° Allow M2 and M3 if axes are reversed Assume origin is (0, 0) if unlabelled. Do not allow M2 or M3 if scale is non-linear or awkward ECF from 1(a)(ii) Allow one plotting error
1(a)(iv)	thin straight line of best fit drawn with even distribution of points ;	1	Line must meet one axis and take into account the point at (5, 15) Allow ECF on points plotted in 1(a)(iii) Do not allow dot to dot drawn with a ruler
1(a)(v)	increasing the (percentage) concentration (of hydrogen peroxide) increases rate (of reaction) ;	1	Allow rate increases with concentration Allow rate is proportional to concentration / it is proportional Allow faster reaction for increased rate Ignore a higher percentage concentration gives a high rate of reaction. Must be comparative Ignore positive relationship / correlation Ignore references to time / temperature
1(a)(vi)	(percentage) concentration (of hydrogen peroxide) ;	1	

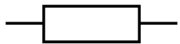
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Question	Answer	Marks	Guidance
1(a)(vii)	to reduce heat loss / reduce thermal energy loss / reduce heat energy transfer to the environment / reduce radiation ; to prevent contamination with enzyme / catalase / yeast / hydrogen peroxide / to prevent changing concentration of hydrogen peroxide / prevent mixing different concentrations of hydrogen peroxide ;	2	Allow insulation Ignore reduces energy loss alone Ignore to reduce heat gain arguments Allow to stop light affecting (rate of) breakdown of hydrogen peroxide Do not allow to keep temperature constant Ignore to prevent temperature loss / decrease Ignore to prevent contamination / remove impurities / chemicals / substances / residues / solutions alone Do not allow to remove acid Ignore references to temperature
1(b)	biuret ; purple ;	2	Mark independently Allow phonetic spelling or errors in order of letters e.g. biuret / buret / burette / buriet Allow lilac / mauve / violet
2(a)	fizzes / bubbles / effervescence ;	1	Allow a blue solution forms Ignore copper carbonate dissolves / disappears / colour changes Ignore gas / carbon dioxide produced / foam produced Ignore named incorrect gases
2(b)	to ensure all the <u>acid</u> is used up / all the <u>acid</u> reacts ;	1	Allow so the acid is fully neutralised Ignore complete reaction / reaction finishes / all the copper carbonate reacts/is used up
2(c)	diagram of correctly assembled filter paper, filter funnel and conical flask ; correct labels (filter) funnel, filter paper ;	2	Note filter needs to be a V shape with a V at the base / no holes in the filter paper or collecting vessel Note funnel must have upper funnel and neck Ignore shape of collecting vessel Allow other collecting vessels e.g. test-tube / beaker Ignore other labels

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Question	Answer	Marks	Guidance
2(d)	(filtrate) blue solution and (residue) green solid ;	1	Must be in the correct order. Allow blue liquid Do not allow green precipitate / powder
2(e)(i)	blue-green ;	1	Allow green-blue Ignore green or blue alone
2(e)(ii)	(blue) flame does not mask the (flame test) <u>colour</u> ;	1	Allow (flame) <u>colour</u> not masked/obscured / (flame) <u>colour</u> is distinct (from colour of Bunsen flame) Allow to see / distinguish / observe / a the <u>colour</u> (clearly) Allow sodium gives a yellow flame

Question	Answer	Marks	Guidance
3	One marking point from each section and any <i>two</i> others (if one section is missing max 6 etc): 1 Apparatus balance (to measure mass of salt) ; measuring cylinder used to measure volume of a liquid added at the start i.e. salt solution or water or detergent ; 2 Method method described clearly such that another student could use it to obtain suitable results, dissolve at least two different amounts of salt in water, add liquid detergent, shake and measure amount of foam / final volume ; 3 Measurements measure <u>mass</u> of salt / gives <u>stated mass(es)</u> of salt <u>with units</u> ; use at least five different <u>masses</u> of salt ; measure the, <u>volume</u> of foam / <u>volume</u> of solution and foam after shaking / (total) volume after shaking ; 4 Control variables same volume of water / volume of salt solution / same volume of liquid detergent ; same temperature <u>of water</u> ; same time of shaking / number of shakes ; 5 Processing results calculate volume of foam by subtracting initial <u>volume</u> from final <u>volume</u> / calculate difference between initial and final <u>volume</u> ; plot a graph of <u>mass</u> of salt against <u>volume</u> of foam ;	7	Ignore scale Allow burette / <u>graduated</u> pipette / graduated cylinder for 'measuring cylinder' Ignore teat pipette / pipette / syringe / volumetric pipette Allow 'amounts' in method section Ignore do experiment with no salt / 0 g salt for method and measurements Ignore 'measure amounts' unqualified Allow measure amount of foam <u>in cm³</u> / measure amount of salt <u>in g</u> (may be awarded from a table heading) Allow 5 values of mass in g in a table Allow measure height of foam Allow record = measure but Ignore 'calculate' the foam in Section 3 Allow 'control the ...' for 'use the same ...' Ignore 'known volume' for 'constant volume' Allow 'amounts' in control variables section Ignore same room temperature Allow reversed axes Allow plot a graph of <u>mass</u> of salt against <u>height</u> of foam

Question	Answer	Marks	Guidance
4(a)	P (circled) and correct symbol for resistor ;	1	Do not allow variable resistor / line through resistor 
4(b)(i)	1.35 (V) ; 0.3(0) (A) ;	2	
4(b)(ii)	4.5 (Ω) ;	1	Allow 4.50 Allow ECF from 4(b)(i) correctly rounded to 2 or 3 sig figs
4(c)(i)	70.7 (cm) ;	1	
4(c)(ii)	clip (is too wide and) covers reading / cannot see measurement / scale / markings / reading ;	1	Allow cannot see numbers Ignore cannot determine exact position (repeats question) Ignore clip too wide alone Ignore cannot see ruler
4(c)(iii)	it is not 50 cm away from other clip / clips not 50 cm apart ;	1	Allow clip / x_1 should be at 20.7 (cm) / clips are (only) 20.7 apart / (ECF on incorrect 4(c)(i)) ; Allow distance between crocodile clips is less than 50 cm / too small Ignore clip / is x_1 in wrong place / is at 50 cm / distance should be 50 cm
4(d)(i)	1.25 (V) and 0.35 (A) ;	1	
4(d)(ii)	3.57 (Ω) ;	1	Allow ECF from 4(d)(i) but must be to 3 sig figs correctly rounded
4(e)(i)	3.75 ;	1	Allow ECF from 4(b)(ii)

Question	Answer	Marks	Guidance
4(e)(ii)	Method 1 calculates percentage difference AND states appropriate conclusion i.e. M1 % ge difference is $(3.75 - 3.57) = 0.18 \div 3.75 \times 100 =$ 4.8 OR $(3.75 - 3.57) = 0.18 \div 3.57 \times 100 =$ 5.0 ; M2 (correct calculation plus statement that) yes / they are within 10% / they are within the limits / they are (considered) equal ;	2	Allow ECF on candidate's values Allow truncated or rounded values answers to one or more sig figs e.g. 4.7, 5 etc. i.e. Allow 2 marks for yes and 4.8 / 5.0 Ignore conclusion without calculation = 0 Other methods for M1 $3.57 / 3.75 \times 100 =$ 95.2...% (which is more than 90%) $3.75 / 3.57 \times 100 =$ 105...% (which is less than 110%) Work out 10% increase on 3.57 e.g. (3.57×1.1) OR $(3.57 \times 0.1 + 3.57) =$ 3.927 (which is more than 3.75) Work out 10% reduction on 3.75 e.g. $3.75 \times 0.9 =$ 3.375 (which is less than 3.57) Works out 10% of 3.75 = 0.375 or 3.57 = 0.357 <u>and</u> states that this is more than the difference between the values (0.18) (so yes)
4(f)	reading / voltage / current will be greater than the scale on either / both meters ;	1	Allow volts / amps / 'it' is greater than the scale Allow voltmeter maximum is 2 V and power supply is 12 V Allow voltage will be greater than 2 V / current will be greater than 1 A ; Ignore voltage or current is 'too high' / higher alone / voltmeter maximum is 2 V alone Ignore measurements / numbers are too high Ignore wire heats up